

**Pre-examination training for SC/ST/OBC candidates under Regular Recruitment
Drive for Junior Associates**

Quantitative Aptitude Set-04

Directions (Q1–Q6)

Study the following data carefully and answer the questions.

The table shows the **total population (males + females)** of four cities in 2020 and 2021.

City	2020	2021
P	360	540
Q	480	850
R	600	660
S	750	450

Q1.

The ratio of males to females in city P in 2020 is **4:5**. The number of females in city P in 2021 is 1.25 times the number of females in 2020. Find the difference between the number of males in city P in 2020 and 2021.

- (a) 125
- (b) 112
- (c) 150
- (d) 155
- (e) 130

Answer: (e) 130

Solution:

Males in P in 2020 = $360 \times \frac{4}{9} = 160$

Females in 2020 = $360 - 160 = 200$

Females in 2021 = $200 \times 1.25 = 250$

Males in 2021 = $540 - 250 = 290$

Required difference = $290 - 160 = \mathbf{130}$

Q2.

The population of city T in 2020 is 35% more than that of city Q in 2020. If the total population of city T in 2020 and 2021 together is 1350, find its population in 2021.

- (a) 656
- (b) 702
- (c) 734

- (d) 808
- (e) 619

Answer: (b) 702

Solution:

Population of T in 2020 = $480 \times 135/100 = 648$

Population of T in 2021 = $1350 - 648 = \mathbf{702}$

Q3.

The population of city Q in 2019 was 30% less than the average population of cities P and R in 2021. Find the ratio of the population of Q in 2019 to that of S in 2020.

- (a) 13:25
- (b) 14:25
- (c) 16:27
- (d) 12:23
- (e) 14:27

Answer: (b) 14:25

Solution:

Average population of P and R in 2021 = $(540 + 660)/2 = 600$

Q's population in 2019 = 70% of 600 = 420

Required ratio = $420 : 750 = \mathbf{14 : 25}$

Q4.

The number of males in city S in 2020 is 40% of the total population of city Q in 2021. The number of males in S in 2021 is half the number of males in S in 2020. Find the total number of females in S in 2020 and 2021 together.

- (a) 690
- (b) 648
- (c) 605
- (d) 569
- (e) 719

Answer: (a) 690

Solution:

Males in S in 2020 = 40% of 850 = 340

Females in S in 2020 = $750 - 340 = 410$

Males in S in 2021 = $340 \times 1/2 = 170$

Females in S in 2021 = $450 - 170 = 280$

Required total = $410 + 280 = \mathbf{690}$

Q5.

The population of city R in 2018 was 60% of the population of city P in 2020. If 25% of the population of R in 2018 were males, find the number of females as a percentage of the population of city S in 2021.

- (a) 31%
- (b) 25%
- (c) 21%
- (d) 36%
- (e) 43%

Answer: (d) 36%

Solution:

Population of R in 2018 = 60% of 360 = 216

Males = 25% of 216 = 54

Females = 216 - 54 = 162

Required percentage = $162/450 \times 100 = 36\%$

Q6.

$4/5$ of the population of city S in 2020 are literate, while $5/12$ of the population of city Q in 2020 are illiterate. Find the difference between the total literate population of Q and S together and the total illiterate population of Q and S together.

- (a) 530
- (b) 622
- (c) 576
- (d) 486
- (e) 449

Answer: (a) 530

Solution:

Literate in S = $750 \times 4/5 = 600$

Illiterate in S = $750 - 600 = 150$

Illiterate in Q = $480 \times 5/12 = 200$

Literate in Q = $480 - 200 = 280$

Total literates = $600 + 280 = 880$

Total illiterates = $150 + 200 = 350$

Difference = $880 - 350 = 530$

Directions (Q7-Q10)

Read the following information carefully.

There are **396 students** in three colleges X, Y and Z.

- Number of girls in Y = 60.
- Boys in X are 70% of girls in Y.
- Girls in X are 18 more than girls in Y.
- Girls in Z are 1.6 times the girls in Y.
- Girls in Z are 3.2 times the boys in Z.
- Boys in Y are 12 more than girls in X.

From the information:

College	Boys	Girls	Total
X	42	78	120
Y	90	60	150
Z	30	96	126
Total	162	234	396

Q7.

Find the difference between boys in X and Y together and girls in Y and Z together.

- (a) 35
- (b) 19
- (c) 32
- (d) 24
- (e) 27

Answer: (d) 24

Solution:

$$\text{Boys in X + Y} = 42 + 90 = 132$$

$$\text{Girls in Y + Z} = 60 + 96 = 156$$

$$\text{Difference} = 24$$

Q8.

Boys in college X are what percentage of the total students in college Z?

- (a) 40.22%
- (b) 33.33%
- (c) 38.17%
- (d) 27.15%
- (e) 29.41%

Answer: (b) 33.33%

Solution:

$$\text{Required percentage} = 42/126 \times 100 = 33.33\% \text{ approximately}$$

Q9.

Find the ratio of girls in X to boys in Z.

- (a) 11:5
- (b) 13:7
- (c) 13:5
- (d) 15:4
- (e) 17:6

Answer: (c) 13:5

Solution:

Girls in X : Boys in Z = $78 : 30 = \mathbf{13 : 5}$

Q10.

The total number of students in college W is 30% more than the total number of students in college X. The ratio of boys in Z to boys in W is 3:4. Find the number of girls in W.

- (a) 131
- (b) 107
- (c) 127
- (d) 102
- (e) 116

Answer: (e) 116

Solution:

Students in W = 130% of 120 = 156

Boys in Z = 30

Boys Z : Boys W = 3:4

Boys in W = $30 \times \frac{4}{3} = 40$

Girls in W = $156 - 40 = \mathbf{116}$

Q11. Mixture

The ratio of milk to water in Jar A is **11:7**, while the ratio in Jar B is **4:9**. The quantity of water in Jar A is 2 litres more than the quantity of water in Jar B. After mixing both jars, the total quantity of milk exceeds the total quantity of water by 2 litres. Find the quantity of water in the final mixture.

- (a) 115 litres
- (b) 110 litres
- (c) 105 litres
- (d) 101 litres
- (e) 120 litres

Answer: (b) 110 litres

Solution:

Let Jar A contain:

$$\text{Milk} = 11x, \text{Water} = 7x$$

Jar B:

$$\text{Milk} = 4y, \text{Water} = 9y$$

Given:

$$7x - 9y = 2 \dots\dots(1)$$

Also, total milk – total water = 2

$$(11x + 4y) - (7x + 9y) = 2$$

$$4x - 5y = 2 \dots\dots(2)$$

$$\text{From (2): } x = (5y + 2)/4$$

$$\text{Substituting in (1): } 7(5y + 2)/4 - 9y = 2 \rightarrow 35y + 14 - 36y = 8 \rightarrow -y + 14 = 8 \rightarrow y = 6$$

$$x = (30 + 2)/4 = 8$$

$$\text{Water in Jar A} = 7 \times 8 = 56 \text{ litres}$$

$$\text{Water in Jar B} = 9 \times 6 = 54 \text{ litres}$$

$$\text{Total water} = 56 + 54 = \mathbf{110 \text{ litres}}$$

Directions (Q12-Q14)

Solve both equations and establish the relationship between x and y.

(a) $x > y$

(b) $x \geq y$

(c) $x < y$

(d) $x \leq y$

(e) $x = y$ or no relation can be established

Q12.

$$\text{I. } 2x^2 - 22x + 48 = 0$$

$$\text{II. } y^2 - 10y + 24 = 0$$

Answer: (e) No relation

Solution:

$$\text{I. } 2x^2 - 22x + 48 = 0 \rightarrow x^2 - 11x + 24 = 0$$

$$(x - 3)(x - 8) = 0$$

$$x = 3, 8$$

$$\text{II. } y^2 - 10y + 24 = 0$$

$$(y - 4)(y - 6) = 0$$

$$y = 4, 6$$

Since some values of x are greater than y while others are smaller: **No definite relationship.**

Q13.

I. $x^2 - 27x + 180 = 0$

II. $3y^2 - 39y + 120 = 0$

Answer: (a) $x > y$

Solution:

I. $(x - 12)(x - 15) = 0$

$x = 12, 15$

II. $3y^2 - 39y + 120 = 0 \rightarrow y^2 - 13y + 40 = 0$

$(y - 5)(y - 8) = 0$

$y = 5, 8$

Every possible value of x is greater than every possible value of y .
Therefore, **$x > y$.**

Q14.

I. $x^2 - 4x - 12 = 0$

II. $y^2 - 15y + 54 = 0$

Answer: (d) $x \leq y$

Solution:

I. $(x + 2)(x - 6) = 0$

$x = -2, 6$

II. $(y - 6)(y - 9) = 0$

$y = 6, 9$

Every possible value of x is less than or equal to every possible value of y .
Therefore: **$x \leq y$**

Directions (Q15-Q18)

Find the approximate value of the question mark (?).

Q15.

$54.90\% \text{ of } 280.10 + 24.95\% \text{ of } 340.20 - 74.80 = ?$

(a) 145

(b) 137

(c) 164

- (d) 191
- (e) 181

Answer: (c) 164

Solution:

Approximate:

$$\begin{aligned} &55\% \text{ of } 280 + 25\% \text{ of } 340 - 75 \\ &= 154 + 85 - 75 \\ &= \mathbf{164} \end{aligned}$$

Q16.

$$88.10 \div 8.02 \times 10.90 + 180.20 = ?^2 + 12.10 \times 10.90$$

- (a) 28
- (b) 4
- (c) 19
- (d) 16
- (e) 13

Answer: (e) 13

Solution:

$$\text{Approximate: } 88 \div 8 \times 11 + 180 = 11 \times 11 + 180 = 121 + 180 = 301$$

$$\text{And, } 12 \times 11 = 132$$

$$\text{Therefore, } ?^2 \approx 301 - 132 = 169$$

$$? \approx \sqrt{169} = \mathbf{13}$$

Q17.

$$?\% \text{ of } (999.70 - 249.60 - 199.80 - 100.30) = 180.30$$

- (a) 43
- (b) 40
- (c) 26
- (d) 32
- (e) 50

Answer: (b) 40

Solution:

$$\text{Approximate expression: } 1000 - 250 - 200 - 100 = 450$$

$$\text{So, } ?\% \text{ of } 450 = 180$$

$$?/100 \times 450 = 180$$

$$? = \mathbf{40}$$

Q18.

$$(? - 420.20) - 45.30\% \text{ of } (2520.40 \div 17.98) = 24.30 \times 15.20 \div 60.60$$

- (a) 489
- (b) 467
- (c) 583
- (d) 525
- (e) 422

Answer: (a) 489

Solution:

Approximate: $2520 \div 18 = 140$

45% of 140 = 63

Right side: $24 \times 15 \div 60 = 6$

Therefore, $? - 420 - 63 = 6$

$? = \mathbf{489}$

Directions (Q19-Q24)

The following table shows the **average number of students visiting Schools A and B** on five different days.

Day	Average of A & B
Monday	61
Tuesday	63
Wednesday	82
Thursday	42
Friday	74

Additional information:

- On Monday, students visiting from School B are **18 more** than those from A.
- The ratio of students visiting from B from Monday to Friday is **7:4:6:3:5**.

Basic Calculation

Since average = $(A+B)/2$:

Day	A+B	A	B
Monday	122	52	70
Tuesday	126	86	40
Wednesday	164	104	60
Thursday	84	54	30
Friday	148	98	50

Q19.

On Monday, the ratio of boys to girls visiting from A is 7:6. The number of boys visiting from B on Friday is equal to the number of girls visiting from A on Monday. Find the girls visiting from B on Friday as a percentage of the students visiting from A on Wednesday.

- (a) 33%
- (b) 39%
- (c) 25%
- (d) 20%
- (e) 16%

Answer: (c) 25%

Solution:

A students on Monday = 52

Boys : Girls = 7:6

Girls from A = $52 \times 6/13 = 24$

Therefore, boys from B on Friday = 24.

Total students from B on Friday = 50.

Girls from B on Friday = $50 - 24 = 26$

Required percentage = $26/104 \times 100 = 25\%$

Q20.

The total number of students in School C is 30% more than the average number of students visiting from B on Monday, Thursday and Friday. The ratio of total students in C to D is 5:7. Find the number of students in D.

- (a) 91
- (b) 78
- (c) 95
- (d) 104
- (e) 86

Answer: (a) 91

Solution:

Average B students = $(70 + 30 + 50)/3 = 150/3 = 50$

Students in C = $130\% \times 50 = 65$

C : D = 5 : 7

D = $65 \times 7/5 = 91$

Q21.

Find the average number of students visiting from School B on all five days.

- (a) 54
- (b) 43
- (c) 50

- (d) 38
- (e) 59

Answer: (c) 50

Solution:

Total B students = $70 + 40 + 60 + 30 + 50 = 250$
Average = $250/5 = 50$

Q22.

Students from A visiting on Wednesday participated in two activities — drama and music — in the ratio 9:4. Find the difference between students participating in music and the total number of students visiting from both schools on Thursday.

- (a) 43
- (b) 47
- (c) 40
- (d) 52
- (e) 62

Answer: (d) 52

Solution:

A students on Wednesday = 104.
Drama : Music = 9 : 4
Music participants = $104 \times 4/13 = 32$
Total A+B students on Thursday = 84
Required difference = $84 - 32 = 52$

Q23.

The number of students visiting from A on Tuesday is what percentage more than the students visiting from B on Friday?

- (a) 60%
- (b) 72%
- (c) 78%
- (d) 66%
- (e) 85%

Answer: (b) 72%

Solution:

A on Tuesday = 86
B on Friday = 50
Difference = 36
Required percentage = $36/50 \times 100 = 72\%$

Q24.

On a particular week, 25% of the students scheduled to visit from A on Monday did not visit, while 30% of those scheduled from B on Wednesday did not visit. Find the total number of students who did not visit.

- (a) 31
- (b) 17
- (c) 46
- (d) 35
- (e) 22

Answer: (a) 31

Solution:

A on Monday = 52.

Non-visitors = 25% of 52 = 13

B on Wednesday = 60.

Non-visitors = 30% of 60 = 18

Total non-visitors = 13 + 18 = **31**

Directions (Q25-Q30)

Study the following data carefully and answer the questions.

The table shows the total population (males + females) of four cities in 2020 and 2021.

City	2020	2021
J	650	880
K	560	900
L	640	560
M	792	650

Q25.

The ratio of males to females in city J in 2020 is 8:5. The number of females in city J in 2021 is 1.3 times the number of females in 2020. Find the difference between the number of males in city J in 2020 and 2021.

- (a) 142
- (b) 155
- (c) 165
- (d) 184
- (e) 177

Answer: (b) 155

Solution:

Males in J in 2020 = $650 \times \frac{8}{13} = 400$
Females in 2020 = $650 - 400 = 250$
Females in 2021 = $250 \times 1.3 = 325$
Males in 2021 = $880 - 325 = 555$
Required difference = $555 - 400 = 155$

Q26.

The population of city N in 2020 is 40% more than that of city K in 2020. If the total population of city N in 2020 and 2021 together is 1750, find its population in 2021.

- (a) 1078
- (b) 798
- (c) 966
- (d) 1136
- (e) 912

Answer: (c) 966

Solution:

Population of N in 2020 = $560 \times \frac{140}{100} = 784$
Population of N in 2021 = $1750 - 784 = 966$

Q27.

The population of city K in 2019 was 35% less than the average population of cities J and L in 2021. Find the ratio of the population of K in 2019 to that of M in 2020.

- (a) 13:22
- (b) 11:22
- (c) 13:20
- (d) 15:23
- (e) 17:26

Answer: (a) 13:22

Solution:

Average population of J and L in 2021 = $(880 + 560)/2 = 720$
K's population in 2019 = 65% of 720 = 468
Required ratio = $468 : 792 = 13 : 22$

Q28.

The number of males in city M in 2020 is 35% of the total population of city K in 2021. The number of males in M in 2021 is three-fifth of the number of males in M in 2020. Find the total number of females in M in 2020 and 2021 together.

- (a) 883
- (b) 820

- (c) 1002
- (d) 938
- (e) 1104

Answer: (d) 938

Solution:

Males in M in 2020 = 35% of 900 = 315
Females in M in 2020 = 792 – 315 = 477
Males in M in 2021 = $315 \times \frac{3}{5} = 189$
Females in M in 2021 = 650 – 189 = 461
Required total = 477 + 461 = 938

Q29.

The population of city L in 2018 was 60% of the population of city J in 2020. If 20% of the population of L in 2018 were males, find the number of females as a percentage of the population of city M in 2021.

- (a) 38%
- (b) 42%
- (c) 34%
- (d) 48%
- (e) 54%

Answer: (d) 48%

Solution:

Population of L in 2018 = 60% of 650 = 390
Males = 20% of 390 = 78
Females = 390 – 78 = 312
Required percentage = $\frac{312}{650} \times 100 = 48\%$

Q30.

$\frac{3}{4}$ of the population of city M in 2020 are literate, while $\frac{3}{8}$ of the population of city K in 2020 are illiterate. Find the difference between the total literate population of K and M together and the total illiterate population of K and M together.

- (a) 536
- (b) 562
- (c) 509
- (d) 487
- (e) 623

Answer: (a) 536

Solution:

Literate in M = $792 \times \frac{3}{4} = 594$
 Illiterate in M = $792 - 594 = 198$
 Illiterate in K = $560 \times \frac{3}{8} = 210$
 Literate in K = $560 - 210 = 350$
 Total literates = $594 + 350 = 944$
 Total illiterates = $198 + 210 = 408$
 Difference = $944 - 408 = 536$

Directions (Q31–Q34)

Read the following information carefully.

There are 335 students in three colleges D, E and F.

- Number of girls in E = 50.
- Boys in D are 60% of girls in E.
- Girls in D are 20 more than girls in E.
- Girls in F are 1.5 times the girls in E.
- Girls in F are 2.5 times the boys in F.
- Boys in E are 10 more than girls in D.

From the information:

College	Boys	Girls	Total
D	30	70	100
E	80	50	130
F	30	75	105
Total	140	195	335

Q31.

Find the difference between boys in D and E together and girls in E and F together.

- (a) 15
 (b) 6
 (c) 12
 (d) 25
 (e) 21

Answer: (a) 15

Solution:

Boys in D + E = $30 + 80 = 110$
 Girls in E + F = $50 + 75 = 125$
 Difference = 15

Q32.

Boys in college D are what percentage of the total students in college F?

- (a) 36.52%
- (b) 33.42%
- (c) 28.57%
- (d) 25.69%
- (e) 22.77%

Answer: (c) 28.57%

Solution:

Required percentage = $30/105 \times 100 = 28.57\%$ approximately

Q33.

Find the ratio of girls in D to boys in F.

- (a) 5:3
- (b) 7:3
- (c) 7:4
- (d) 9:4
- (e) 8:3

Answer: (b) 7:3

Solution:

Girls in D : Boys in F = $70 : 30 = 7 : 3$

Q34.

The total number of students in college G is 15% more than the total number of students in college D. The ratio of boys in F to boys in G is 3:4. Find the number of girls in G.

- (a) 75
- (b) 60
- (c) 85
- (d) 88
- (e) 81

Answer: (a) 75

Solution:

Students in G = $115\% \times 100 = 115$

Boys in F = 30.

Boys F : Boys G = 3:4

Boys in G = $30 \times 4/3 = 40$

Girls in G = $115 - 40 = 75$

Q35. Mixture

The ratio of milk to water in Jar A is 5:2, while the ratio in Jar B is 7:4. The quantity of milk in Jar A is 8 litres more than the quantity of milk in Jar B. After mixing both jars, the total quantity of milk exceeds the total quantity of water by 48 litres. Find the quantity of milk in the final mixture.

- (a) 92 litres
- (b) 83 litres
- (c) 97 litres
- (d) 100 litres
- (e) 104 litres

Answer: (a) 92 litres

Solution:

Let Jar A contain:

Milk = $5x$, Water = $2x$

Jar B:

Milk = $7y$, Water = $4y$

Given:

$$5x - 7y = 8 \dots\dots(1)$$

Also, total milk – total water = 48

$$(5x + 7y) - (2x + 4y) = 48$$

$$3x + 3y = 48$$

$$x + y = 16 \dots\dots(2)$$

$$\text{So, } y = 16 - x$$

Substituting in (1): $5x - 7(16 - x) = 8$

$$5x - 112 + 7x = 8$$

$$12x = 120$$

$$x = 10$$

$$y = 16 - 10 = 6$$

Milk in Jar A = $5 \times 10 = 50$ litres

Milk in Jar B = $7 \times 6 = 42$ litres

Total milk = $50 + 42 = 92$ litres

Directions (Q36-Q38)

Solve both equations and establish the relationship between x and y .

- (a) $x > y$
- (b) $x \geq y$
- (c) $x < y$
- (d) $x \leq y$
- (e) $x = y$ or no relation can be established

Q36.

I. $4x^2 - 40x + 36 = 0$

II. $y^2 - 7y + 10 = 0$

Answer: (e) No relation

Solution:

I. $4x^2 - 40x + 36 = 0 \rightarrow x^2 - 10x + 9 = 0$

$(x - 1)(x - 9) = 0$

$x = 1, 9$

II. $y^2 - 7y + 10 = 0$

$(y - 2)(y - 5) = 0$

$y = 2, 5$

Since some values of x are greater than y while others are smaller: No definite relationship.

Q37.

I. $x^2 - 17x + 70 = 0$

II. $y^2 - 10y + 21 = 0$

Answer: (b) $x \geq y$

Solution:

I. $(x - 7)(x - 10) = 0$

$x = 7, 10$

II. $(y - 3)(y - 7) = 0$

$y = 3, 7$

Every possible value of x is greater than or equal to every possible value of y.

Therefore: $x \geq y$

Q38.

I. $x^2 - 6x + 8 = 0$

II. $2y^2 - 30y + 108 = 0$

Answer: (c) $x < y$

Solution:

I. $(x - 2)(x - 4) = 0$

$x = 2, 4$

II. $2y^2 - 30y + 108 = 0 \rightarrow y^2 - 15y + 54 = 0$

$(y - 6)(y - 9) = 0$

$y = 6, 9$

Every possible value of x is less than every possible value of y .
Therefore, $x < y$.

Directions (Q39–Q42)

Find the approximate value of the question mark (?).

Q39.

$$64.90\% \text{ of } 320.10 + 34.95\% \text{ of } 180.20 - 89.80 = ?$$

- (a) 169
- (b) 194
- (c) 181
- (d) 207
- (e) 216

Answer: (c) 181

Solution:

Approximate:

$$\begin{aligned} &65\% \text{ of } 320 + 35\% \text{ of } 180 - 90 \\ &= 208 + 63 - 90 \\ &= 181 \end{aligned}$$

Q40.

$$98.10 \div 7.02 \times 14.10 + 200.20 = ?^2 + 14.10 \times 10.05$$

- (a) 13
- (b) 16
- (c) 27
- (d) 22
- (e) 9

Answer: (b) 16

Solution:

$$\begin{aligned} &\text{Approximate: } 98 \div 7 \times 14 + 200 = 14 \times 14 + 200 = 196 + 200 = 396 \\ &\text{And, } 14 \times 10 = 140 \\ &\text{Therefore, } ?^2 \approx 396 - 140 = 256 \\ &? \approx \sqrt{256} = 16 \end{aligned}$$

Q41.

$$34.90\% \text{ of } (1199.60 - 299.70 - 199.90 - 140.00) = ?$$

- (a) 196
- (b) 225
- (c) 170

- (d) 217
(e) 208

Answer: (a) 196

Solution:

Approximate expression: $1200 - 300 - 200 - 140 = 560$

So, $? = 35\%$ of 560

$? = 35/100 \times 560$

$? = 196$

Q42.

$(? - 350.20) - 30.10\% \text{ of } (3600.40 \div 19.98) = 18.20 \times 20.10 \div 72.60$

- (a) 482
(b) 366
(c) 437
(d) 409
(e) 344

Answer: (d) 409

Solution:

Approximate: $3600 \div 20 = 180$

30% of 180 = 54

Right side: $18 \times 20 \div 72 = 5$

Therefore, $? - 350 - 54 = 5$

$? = 409$

Directions (Q43-Q48)

The following table shows the average number of students visiting Schools A and B on five different days.

Day	Average of A & B
Monday	105
Tuesday	120
Wednesday	115
Thursday	75
Friday	135

Additional information:

- On Monday, students visiting from School B are 60 more than those from A.
- The ratio of students visiting from B from Monday to Friday is 9:5:7:4:6.

Basic Calculation

Since average = $(A+B)/2$:

Day	A+B	A	B
Monday	210	75	135
Tuesday	240	165	75
Wednesday	230	125	105
Thursday	150	90	60
Friday	270	180	90

Q43.

On Monday, the ratio of boys to girls visiting from A is 7:8. The number of boys visiting from B on Friday is equal to the number of girls visiting from A on Monday. Find the girls visiting from B on Friday as a percentage of the students visiting from A on Wednesday.

- (a) 40%
- (b) 36%
- (c) 25%
- (d) 53%
- (e) 30%

Answer: (a) 40%

Solution:

A students on Monday = 75

Boys : Girls = 7:8

Girls from A = $75 \times 8/15 = 40$

Therefore, boys from B on Friday = 40.

Total students from B on Friday = 90.

Girls from B on Friday = $90 - 40 = 50$

Required percentage = $50/125 \times 100 = 40\%$

Q44.

The total number of students in School G is 40% more than the average number of students visiting from B on Monday, Thursday and Friday. The ratio of total students in G to H is 7:6. Find the number of students in H.

- (a) 123
- (b) 114
- (c) 96
- (d) 118
- (e) 101

Answer: (b) 114

Solution:

$$\text{Average B students} = (135 + 60 + 90)/3 = 285/3 = 95$$

$$\text{Students in G} = 140\% \times 95 = 133$$

$$G : H = 7 : 6$$

$$H = 133 \times 6/7 = 114$$

Q45.

Find the average number of students visiting from School B on all five days.

(a) 104

(b) 83

(c) 93

(d) 89

(e) 98

Answer: (c) 93

Solution:

$$\text{Total B students} = 135 + 75 + 105 + 60 + 90 = 465$$

$$\text{Average} = 465/5 = 93$$

Q46.

Students from A visiting on Wednesday participated in two activities — drama and music — in the ratio 3:2. Find the difference between students participating in music and the total number of students visiting from both schools on Thursday.

(a) 116

(b) 110

(c) 89

(d) 100

(e) 104

Answer: (d) 100

Solution:

$$\text{A students on Wednesday} = 125.$$

$$\text{Drama : Music} = 3 : 2$$

$$\text{Music participants} = 125 \times 2/5 = 50$$

$$\text{Total A+B students on Thursday} = 150$$

$$\text{Required difference} = 150 - 50 = 100$$

Q47.

The number of students visiting from A on Tuesday is what percentage more than the students visiting from B on Friday?

- (a) 78.8%
- (b) 76.08%
- (c) 86.01%
- (d) 83.33%
- (e) 91.06%

Answer: (d) 83.33%

Solution:

A on Tuesday = 165

B on Friday = 90

Difference = 75

Required percentage = $75/90 \times 100 = 83.33\%$

Q48.

On a particular week, 20% of the students scheduled to visit from A on Monday did not visit, while 40% of those scheduled from B on Wednesday did not visit. Find the total number of students who did not visit.

- (a) 57
- (b) 62
- (c) 67
- (d) 52
- (e) 47

Answer: (a) 57

Solution:

A on Monday = 75.

Non-visitors = 20% of 75 = 15

B on Wednesday = 105.

Non-visitors = 40% of 105 = 42

Total non-visitors = 15 + 42 = 57